

This exam has 2 pages and 8 exercises.

The result will be computed as (total number of points plus 10) divided by 10.

1. Island or liars and truth speakers (11 points)

Suppose you ask the following question to an islander hurrying by on a T-crossing to find the way to the harbour:

Is the declarative sentence “*you are a truth speaker* $\leftrightarrow$ *the harbour is to the right*” true?

Either explain that this question suffices to determine the way to the harbour, or explain why this is not the case.

2. DPLL procedure (11 points)

Apply the DPLL procedure to the CNF

$$(p \vee \neg q \vee \neg r) \wedge (\neg p \vee q \vee \neg r) \wedge (\neg p \vee \neg q \vee \neg r) \wedge r$$

to check whether it is satisfiable. Explain in detail how the procedure comes to a result. Let the procedure branch first on p , then on q , and then on r ; and let it always try the truth value true ($\top$) before false ($\perp$).

3. OBDD (3+9 points)

- (a) Represent $(x \wedge y) \oplus z$ by means of a binary decision tree, with respect to the variable ordering x, y, z .
- (b) Reduce this binary decision tree to an ordered binary decision diagram. (Also give the intermediate reduction steps.)

4. Predicate logic (11 points)

Give two models to show that the formulas $\forall x (C(x) \rightarrow D(x))$ and $\forall x (D(x) \rightarrow C(x))$ do not semantically entail each other. (Explain your answer.)

5. Sets (6 points)

Assuming that $\#U = 16$, $\#A = 6$, $\#B = 8$, $\#C = 7$, $\#(A \cap C) = 3$, $\#(A \cup B' \cup C) = 13$, compute the number of elements in $(A \cup C) \setminus B$.

6. Ordering+Equivalence Relations (3+6+4+5 points)

In the set $V = \{a, b, c, d, e\}$ consider the relation R with matrix representation

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

- a) List all elements of the relation R .
- b) Show that R is an equivalence relation.
- c) Prove that R is not an ordering relation and not a function.
- d) Write down the equivalence classes and a system of representatives for R .

7. Functions (6+4 points)

We are given the functions $f : \mathbb{R} \rightarrow \mathbb{R}$, $g : \mathbb{R} \rightarrow \mathbb{R}$, $mul_2 : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \sqrt{|x| - 1}, \quad g(x) = \frac{1}{x^2 + 1}, \quad \text{and} \quad mul_2(x) = 2x.$$

- a) For each function, determine its domain of definition and its range.
- b) Give a description (and expression for $h(x)$) for $h = g \circ mul_2 \circ f$ and check whether h is injective.

8. Induction and recursion (3+8 points)

We claim that the following statement is true for all natural numbers $n \geq 1$:

$$\sum_{k=1}^n 2(k-2) = n^2 - 3n.$$

- a) Verify by explicit computation that the claim is true for $n = 1$, $n = 2$ and $n = 3$.
- b) Prove by mathematical induction that the statement holds true for all natural numbers n .